Standard Operating Procedure (SOP) for Ericsson PE (YEL-PE-11) - Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on Yelahanka Sub-station

Node: YEL-PE-11 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_3881

Description

This Standard Operating Procedure (SOP) outlines the steps to troubleshoot and resolve a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on the Ericsson PE (YEL-PE-11) at the Yelahanka Sub-station. The Routing Protocol Flap occurs when there is an excessive and rapid change in the routing information exchanged between the PE and its neighboring routers. This can be due to various reasons such as misconfiguration, hardware failure, or software issues. The BGP route churn occurs when there is a rapid change in the routing information exchanged between the PE and its neighboring BGP routers, while the OSPF adjacency loss occurs when there is an issue with the OSPF neighbor relationship between the PE and its neighboring OSPF routers. The ISIS LSP flooding occurs when there is an excessive and rapid change in the ISIS routing information exchanged between the PE and its neighboring ISIS routers. In this SOP, we will focus on identifying the root cause of the Routing Protocol Flap and resolving it to ensure stable and reliable routing operations.

Pre-requisite Checks

- pre-requisite: Verify the site location and node ID (YEL-PE-11) to ensure the correct equipment is being targeted., procedure: Use the command "show location" to verify the site location and "show node" to verify the node ID.

- pre-requisite: Check the redundant link status to ensure there are no issues with the redundant links., procedure: Use the command "show link redundancy" to verify the redundant link status.

- pre-requisite: Verify the software version and configuration of the PE to ensure it is up-to-date and correctly configured., procedure: Use the command "show version" to verify the software version and "show configuration" to verify the configuration.

Potential Root Causes

Cause	Probability
BGP configuration issue	High
OSPF neighbor relationship issue	Medium
ISIS routing information issue	Low

Diagnostic Commands & Expected Outputs

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Command: show bgp summary, Expected Output: BGP routing information is not being exchanged between the PE and its neighboring BGP routers., Description: This command will display a summary of the BGP routing information exchanged between the PE and its neighboring BGP routers. If there is no BGP routing information being exchanged, it may indicate a BGP configuration issue.

Command: show ospf neighbor, Expected Output: The OSPF neighbor relationship is not established between the PE and its neighboring OSPF routers., Description: This command will display a list of the OSPF neighbors of the PE. If there are no OSPF neighbors or the OSPF neighbor relationship is not established, it may indicate an issue with the OSPF neighbor relationship.

Command: show isis lsp flooding, Expected Output: There is excessive and rapid change in the ISIS routing information exchanged between the PE and its neighboring ISIS routers., Description: This command will display a list of the ISIS LSPs (Link-State Packets) being flooded between the PE and its neighboring ISIS routers. If there is excessive and rapid change in the ISIS routing information, it may indicate an issue with the ISIS routing information.
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Resolution Steps

Step 1: Step: Verify and correct BGP configuration, Command: show bgp configuration, Description: Verify the BGP configuration of the PE and correct any issues that may be causing the BGP routing information issue. Use the command "show bgp configuration" to verify the BGP configuration.

Step 2: Step: Verify and establish OSPF neighbor relationship, Command: show ospf neighbor establishment, Description: Verify the OSPF neighbor relationship between the PE and its neighboring OSPF routers and establish the relationship if it is not established. Use the command "show ospf neighbor establishment" to verify the OSPF neighbor relationship.

Step 3: Step: Verify and correct ISIS routing information, Command: show isis lsp verification, Description: Verify the ISIS routing information exchanged between the PE and its neighboring ISIS routers and correct any issues that may be causing the ISIS LSP flooding. Use the command "show isis lsp verification" to verify the ISIS routing information.

Post-Check Verification

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Command: show bgp summary, Expected Output: BGP routing information is being exchanged between the PE and its neighboring BGP routers., Description: Verify that the BGP routing information is being exchanged between the PE and its neighboring BGP routers.

Command: show ospf neighbor, Expected Output: The OSPF neighbor relationship is established between the PE and its neighboring OSPF routers., Description: Verify that the OSPF neighbor relationship is established between the PE and its neighboring OSPF routers.

Command: show isis lsp flooding, Expected Output: There is no excessive and rapid change in the ISIS routing information exchanged between the PE and its neighboring ISIS routers., Description: Verify that there is no excessive and rapid change in the ISIS routing information exchanged between the PE and its neighboring ISIS routers.
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Roll-back Procedures

- Step: Restore previous BGP configuration, Command: apply configuration backup, Description: Restore the previous BGP configuration if the fix fails.

- Step: Restore previous OSPF neighbor relationship, Command: apply neighbor relationship backup, Description: Restore the previous OSPF neighbor relationship if the fix fails.

- Step: Restore previous ISIS routing information, Command: apply isis lsp backup, Description: Restore the previous ISIS routing information if the fix fails.